8 The Sun consists largely of hydrogen, ${}_{1}^{1}\mathrm{H}$ (a proton) and helium, ${}_{2}^{4}\mathrm{He}$. At its core, temperature reaches a very high value. A continuous process of nuclear fusion occurs, releasing immense amount of energy. This energy is then radiated by the Sun.

One of the primary reaction pathways responsible for this energy generation is the **proton-proton cycle** (also known as the **hydrogen cycle**). The sequence of nuclear reactions involved in this cycle is detailed below:

$${}_{1}^{1}\mathrm{H} + {}_{1}^{1}\mathrm{H} \rightarrow {}_{1}^{2}\mathrm{H} + {}_{+1}^{0}\mathrm{e} + \mathrm{X} \quad \text{---------- equation (1)}$$

$${}_{1}^{2}\mathrm{H} + {}_{1}^{1}\mathrm{H} \rightarrow {}_{2}^{3}\mathrm{He} \quad \text{---------- equation (2)}$$

$${}_{2}^{3}\mathrm{He} + {}_{2}^{3}\mathrm{He} \rightarrow {}_{2}^{4}\mathrm{He} + 2({}_{1}^{1}\mathrm{H}) \quad \text{---------- equation (3)}$$

In each of these reactions, energy is released. The reactions of this cycle give us most of the energy which we receive from the Sun.

In a nuclear fusion, two small charged nuclei come together to form a larger one. From what we already know about the electric potential energy between charged particles, we can work out the conditions necessary for the fusion. From the kinetic theory of gases, the temperature T of the gas and the mean kinetic energy E_k of the particles of the gas like ${}_{1}^{1}\mathrm{H}$ is given by:

$$E_k = \frac{3}{2}kT$$

where k is the Boltzmann's constant.

The Sun radiates energy into space at a constant rate of 3.8×10^{26} W. This power output is sustained entirely by nuclear fusion reactions occurring within the solar core, where protons fuse to form helium.

The Sun has a total mass of 2.0×10^{30} kg. However, the high temperature required for nuclear fusion exist only in its core. As a result, the hydrogen (protons) available for fusion in the core accounts for only 10% of the Sun's total mass.

In the core's fusion cycle, the average net energy released *per proton* consumed is 6.68 MeV.

You may use relevant information in Fig. 8.1 to answer the following questions

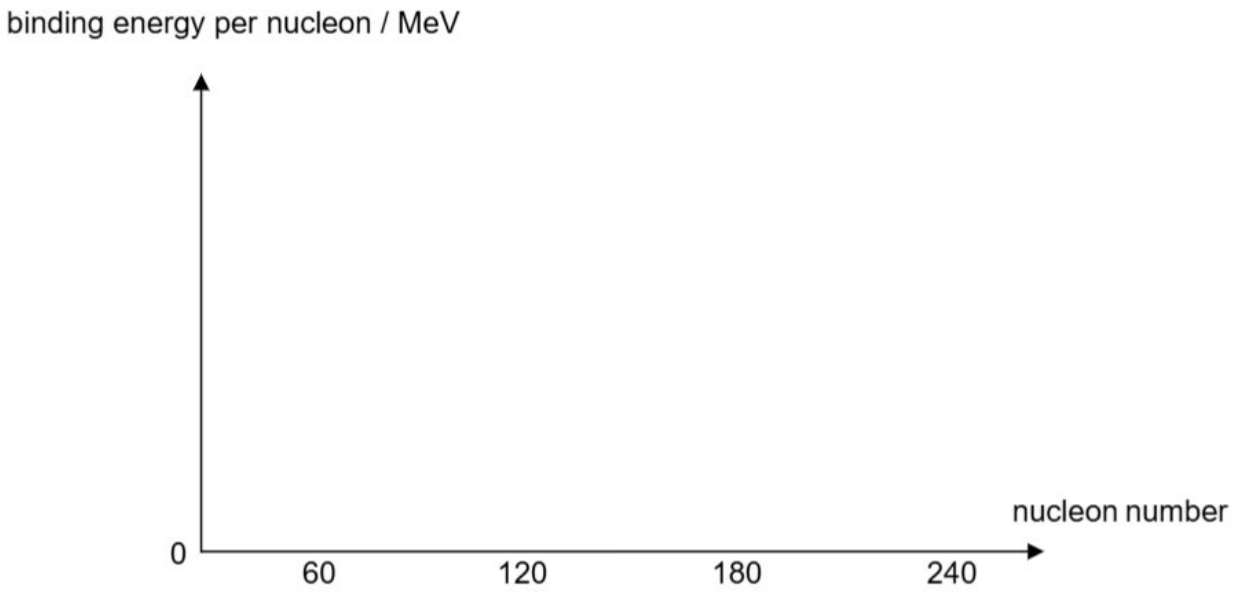
Nuclide	Mass / *u*
$^{0}_{1}e$	0.000 549
$^{1}_{1}H$	1.007 825
$^{1}_{0}n$	1.008 665
$^{2}_{1}H$	2.014 102
$^{3}_{1}H$	3.016 049
$^{3}_{2}He$	3.016 030
$^{4}_{2}He$	4.002 604

Fig. 8.1

(a) **(i)** Determine the binding energy per nucleon of the helium $^{4}_{2}He$ nuclide.

binding energy per nucleon = ... MeV [2]

(ii) Sketch the binding energy per nucleon with for nucleon number from $A = 1$ to $A = 240$.



[1]

(iii) Use your answer in **(a)(i)** and **(a)(ii)** to explain why nuclear fusion may release energy.

.....

..... [1]

(iv) Show that the energy released for equation (2) is 5.5 MeV.

(b) In the first reaction, shown in equation (1), it is observed that the particle $^{0}_{+1}e$, a positron, is emitted with varying amounts of kinetic energy.

Use relevant conservation law(s) to suggest why emission of a positron in equation (1) could predict the existence of another particle X.

.....

.....

.....

.....

..... [2]

(c) As one proton approaches the other, electrostatic repulsion causes it to slow down, converting their kinetic energy into electric potential energy. For nuclear fusion to take place, the protons must possess enough kinetic energy to reach a separation of 1.0×10^{-14} m where the attractive strong nuclear force becomes dominant.

To fuse two protons together in this way, the temperature must be raised. Using kinetic theory of gases, estimate the minimum temperature necessary for protons to achieve this fusion condition.

temperature = ... K [2]

(d) With reference to equation (1), (2) and (3), determine the net number of protons that are required to produce a helium $^{4}_{2}He$ nucleus.

number = ... [1]

(e) **(i)** Show that there are 1.2×10^{56} protons available for nuclear fusion in the Sun's core.

[1]

(ii) Use **(e)(i)** to estimate the lifetime of the Sun. Give your answer in years.

lifetime of the Sun = ... years [1]